IMPLEMENTATION PLAYBOOK / 01

A practical workflow for two people

ZedBoard Sensor Arbitration Project

Team: Person A + Person B **Plan date:** 29 October 2026
Deadline: 20 November 2026 **Working-demo target:** 15 November

Hardware: PMOD ACL2, PMOD HYGRO, PMOD OLED and ZedBoard. This plan adapts the earlier architecture to two people and reuses your friend's byte-level SPI engine. It defines what to write, who owns it, how blocks connect, and what must pass before integration.

Recommended division

Person A: acquisition/display infrastructure	Person B: scheduling/I2C/software
Existing SPI engine, ACL2 service and device model	I2C engine, HYGRO service and device model
Common modules, record buffering and OLED output	Priority/aging arbiter and transaction dispatcher
Board wiring, top-level integration and counters	AXI interface, small PS UART application and log analysis
Writes tests for A-owned blocks; reviews B's interfaces	Writes tests for B-owned blocks; reviews A's interfaces

Replace A and B with your names in the README. Ownership means one person implements and maintains that file; both review the interfaces. Neither person should wait for the other's sensor hardware before starting: use synthetic requests and records.

Freeze these decisions in your first session

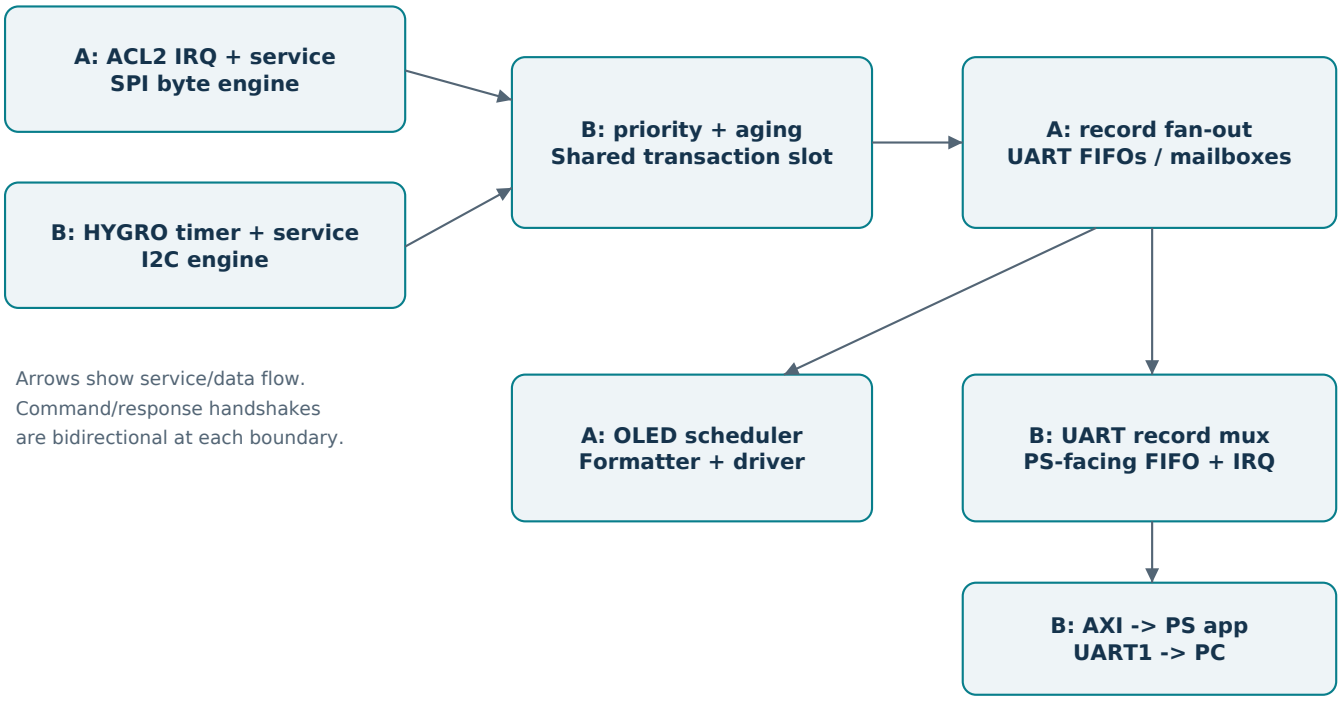
- **HYGRO uses timer events.** HDC1080 has no DRDYn/data-ready pin. Trigger conversion, release the bus slot, wait on a timer, then request a result read. [2,5]
- **Acquisition stays in PL.** The proposed baseline uses a small bare-metal PS bridge only for the onboard USB-UART, which is connected to PS UART1. No Linux is needed. [6]
- **If PL-only is a strict course requirement:** use a PL UART and an external 3.3 V USB-to-TTL adapter instead; omit the AXI/PS bridge files. Confirm this constraint before starting the UART branch.
- **Separate acquisition and output ownership.** Do not hold the sensor grant while waiting for OLED or UART output. Use buffers and separate output grants.

What counts as completion

Live accelerometer and temperature/humidity data, intact UART records, stable OLED text, configurable priorities and aging, a self-checking stress test, and measured fixed-priority versus aging results. This is an implementation plan, not a claim that the listed modules or hardware tests are already complete.

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Architecture and ownership boundaries



Acquisition path: the resource being arbitrated

ACL2 and HYGRO have separate physical buses. To make service arbitration explicit, both compete for a single logical transaction slot. The dispatcher starts either an ACL2 SPI transaction or a HYGRO I2C transaction. Each accepted transaction is non-preemptive and has a timeout. Initial configuration transactions must also use this dispatcher or run in a defined boot-only exclusive phase.

HYGRO conversion is not a bus transaction: its controller relinquishes the slot after triggering conversion. It becomes eligible again after the conversion timer expires. Age only pending, eligible work. This allows an ACL2 read while HYGRO is converting.

Output path: independent consumers

Fan-out sends each completed record to its source UART FIFO and its source OLED latest-value mailbox independently. A full UART queue follows an explicit drop-new-and-count policy, rather than blocking acquisition forever. OLED may replace an older undisplayed value; count this as display coalescing, not as a failed sensor transaction.

The UART scheduler selects a whole 256-bit record for the PS-facing FIFO. The PS is the only physical UART writer and sends one complete formatted line at a time. The OLED scheduler updates a back buffer; the front buffer is immutable throughout a display transfer. Use separate SPI engine instances for ACL2 and OLED.

One clock and one agreed record format

Use a single PL control clock, proposed 100 MHz, for AXI, FIFOs and FSMs. Generate serial timing with clock enables, not divided fabric clocks. Reuse the common-folder record layout from the supplied package. The two developers must not invent separate record or reset conventions.

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Folder structure and reuse rules

```
zedboard-arbiter/
  README.md
  rtl/
    common/       config_pkg.sv, reset_sync.sv,
                  sync_input.sv, sync_fifo.sv
    bus/          spi_master.v, i2c_master.sv
    sensors/      acl2_service.sv, hygro_service.sv
    control/      event_pending.sv,
                  priority_aging_arbiter.sv,
                  transaction_dispatch.sv
    output/       record_fanout.sv, uart_record_mux.sv,
                  oled_update_mux.sv, value_format.sv,
                  oled_driver.sv, font_rom.sv
    metrics/      metrics.sv, traffic_gen.sv
    axi/          axi_regs.sv
    sensor_hub.sv
    zedboard_top.sv
  assets/         font8x8.mem
  constraints/    zedboard_pmods.xdc
  sim/            unit tests, device models, system stress TB
  sw/             main.c, pl_regs.h, pl_irq.c, pl_irq.h
  scripts/        project/build/simulation/log-analysis tools
  docs/           interfaces.md, wiring.md, register_map.md,
                  test_plan.md, demo.md
  results/        raw logs, CSV, plots, run_manifest.csv
  build/          generated tool files (ignored by Git)
```

Use the work already available

Available item	Action
Your friend's spi.v	Keep the module spi_master. Copy to rtl/bus/spi_master.v; do not add another module with the same name.
Your friend's spi_tb.v	Keep as sim/tb_spi_master.sv. Extend it beyond MOSI-to-MISO loopback; do not treat loopback as device verification.
Previously supplied common ZIP	Reuse its four common modules and tb_common.sv. Run the common tests locally in Vivado before integrating.
Different earlier names	adxl362_ctrl = acl2_service; hdc1080_ctrl = hygro_service. Pick the names above and use them consistently.

Compilation and version-control rules

Compile config_pkg.sv before users of its types. Add new sources as SystemVerilog; the existing .v SPI module may remain Verilog. Only one spi_master definition may be compiled. Store tool version and build instructions in README. Commit source, XDC and Tcl; regenerate block-design wrappers, BSP, bitstream, ELF and implementation caches instead of editing generated files.

Status labels: common code = available, locally recheck; SPI byte engine = supplied, needs extended/device tests; every other module in this plan = to implement unless your team already has a verified version.

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Required RTL files: purpose and owner

File under rtl/	Why it is required	By
common/config_pkg.sv	Shared timing defaults, command/response types, IDs, record format and proposed register offsets.	A
common/reset_sync.sv	Asynchronous reset assertion with synchronized release. One instance per clock domain.	A
common/sync_input.sv	Synchronizes ACL2 interrupt and asynchronous control levels; does not capture arbitrary narrow pulses.	A
common/sync_fifo.sv	Small single-clock queues with ready/valid handshakes, occupancy and defined full behavior.	A
bus/spi_master.v	Existing one-byte mode-0 engine; instantiate for ACL2 and separately for OLED. CS belongs to the higher-level controller.	A
bus/i2c_master.sv	START/STOP, address/data, ACK/NACK, open-drain line handling and bounded error termination.	B
sensors/acl2_service.sv	Identity/configuration, CS sequencing, interrupt service and six-byte XYZ burst read. Wraps the byte SPI engine.	A
sensors/hygro_service.sv	Identity/configuration, periodic trigger, conversion timer, bounded retry and four-byte result read. No DRDYn input.	B
control/event_pending.sv	Latches ACL2 requests and timer-generated HYGRO work; records timestamp and repeated/coalesced events. Set wins over simultaneous clear.	B
control/priority_aging_arbiter.sv	Reusable two-requester policy: fixed priority, overdue-request override, oldest-first selection and rotating tie-break.	B
control/transaction_dispatch.sv	Owns one bus transaction at a time; routes command/result to the correct service FSM and releases ownership on success/error/timeout.	B
sensor_hub.sv	Portable integration of sensor/control/output logic. Runs in simulation without the Zynq PS.	A
zedboard_top.sv	Board-specific wrapper around sensor hub and Zynq block design; connects actual PMOD I/O and reset.	A

Important SPI integration detail

The existing SPI engine has no CS output and transfers only one byte per start pulse. This is valid if acl2_service holds CS low over command + address + all data bytes. Give the dispatcher exclusive ownership while that sequence runs; the service FSM starts successive bytes only when the byte engine is idle. Return one response for the complete sensor transaction, not for every byte.

HYGRO transactions and commands must obey the shared package contract. Reading conversion results must not accidentally write a new pointer that restarts conversion. The HYGRO controller is responsible for distinguishing conversion trigger, timed wait and raw result read.

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Required output, measurement and PS files

File	Why it is required	By
rtl/output/record_fanout.sv	Routes completed records independently to UART FIFOs and OLED mailboxes. Counts dropped UART records and overwritten pending display values.	A
rtl/output/uart_record_mux.sv	Uses the reusable arbiter to transfer a complete record from one source FIFO into the PS-facing FIFO. No mixed-source record words.	B
rtl/output/oled_update_mux.sv	Uses a separate arbiter instance to choose a dirty source mailbox and update its region in the back buffer.	A
rtl/output/value_format.sv	Converts signed XYZ and unsigned T/RH into fixed-point text/glyph indices; no floating-point RTL needed.	A
rtl/output/oled_driver.sv	Power/reset/init sequence and complete 512-byte front-buffer transfers over dedicated SPI.	A
rtl/output/font_rom.sv; assets/font8x8.mem	Character bitmap lookup for a simple 16-column x 4-row display.	A
rtl/metrics/metrics.sv	Per-source request/completion/error counts, wait sums/maxima, drops and coherent snapshots.	A
rtl/metrics/traffic_gen.sv	Repeatable synthetic request patterns with bounded mock service times; never drives fake traffic onto real sensor pins.	B
rtl/axi/axi_regs.sv	AXI4-Lite configuration, stable FIFO-head peek, explicit pop, interrupt status and counter snapshots.	B
sw/main.c; sw/pl_regs.h	Bare-metal PS initialization, project register definitions, record formatting and complete UART lines.	B
sw/pl_irq.c; sw/pl_irq.h	GIC/PL interrupt handling and bounded copy from PL FIFO into a software ring; records software drops.	B

Baseline UART: keep sensor control independent

Use PS M_AXI_GP0 -> AXI interconnect -> custom axi_regs, with a PL level interrupt connected to IRQ_F2P. Use a matching Vivado/Vitis or older SDK flow. IRQ remains asserted while records are queued or an enabled sticky error exists. The ISR copies complete records; the main application sends text through PS UART1. Do not format long strings or wait for serial transmission inside the ISR.

Reading record words does not pop. Software reads all eight words from a stable head, then writes FIFO_POP exactly once. Only this software consumer may pop. Record an error on empty pop. Snapshot multiword counters before software reads them.

If strict PL-only is required

Replace axi_regs and all sw/ files with rtl/output/uart_tx.sv and an ASCII message/FIFO path, plus tb_uart_tx.sv. Route PL TX through a PL PMOD pin to the external 3.3 V adapter and connect ground. Both branches use the same record format and arbitration; implement only the selected branch. The built-in USB-UART is not directly connected to PL fabric pins. [6]

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Interfaces and behavior to freeze together

Boundary	Signals / data	Rule
Sensor -> dispatcher	cmd_valid/ready + sensor_cmd_t	Hold command stable until accepted. At most one outstanding transaction per sensor.
Dispatcher -> sensor	rsp_valid/ready + sensor_rsp_t	Exactly one response per accepted transaction, including errors. Source ID must match.
Sensor -> fan-out	record_valid/ready + sensor_record_t	256-bit successful record; hold while stalled. Fan-out commit must be bounded with explicit overflow policy.
Event -> arbiter	pending, eligible, base priorities, aging enable	Only pending AND eligible work competes. Reset wait age on accepted grant, not merely combinational selection.
FIFO -> consumer	valid, ready, data, occupancy	Transfer only on valid AND ready. Data is stable while stalled; no empty pop.
OLED update -> transfer	back buffer, snapshot request, idle/done	Freeze front buffer at transfer start; never modify it until transfer finishes.

Use the exact existing 256-bit record

W0: source ID [7:0], status [15:8], upper bits zero. **W1:** sequence number. **W2:** event timestamp. **W3:** acquisition completion timestamp. **W4:** signed X and Y. **W5:** signed Z, upper half zero. **W6:** unsigned raw temperature and humidity. **W7:** accumulated eligible transaction wait cycles. Zero unused sensor fields. W0 occupies bits [31:0].

Simple aging policy

Normally select the highest base priority. With aging enabled, any eligible request that reaches AGE_LIMIT becomes overdue. Select the oldest overdue request first; break equal-age ties round-robin. Use saturating counters. Never preempt an accepted physical transaction. A timeout must release the transaction slot safely. This avoids ambiguous capped additive-priority behavior.

Configuration and proposed AXI offsets

Preserve common/config_pkg.sv: CONTROL 0x00, PRIORITY 0x04, AGE_LIMIT 0x08, HYGRO_PERIOD 0x0C, STATUS 0x10, IRQ_ENABLE 0x14, IRQ_STATUS 0x18, ERROR_ACK 0x1C, FIFO_LEVEL 0x20, FIFO_POP 0x24, METRICS_CMD 0x28, RECORD_W0..W7 at 0x40..0x5C and snapshot metrics from 0x80. These are project offsets, not chip register addresses. Freeze bit definitions in docs/register_map.md.

Use an AXI slave template with independently accepted AW/W channels, byte strobes, held responses and single-execution side effects. Apply run-mode changes while idle/stopped. Document default priorities, AGE_LIMIT, timeouts, overflow behavior and reset semantics before both people code against them.

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Two-person schedule: parallel tasks and handoffs

Dates	Person A	Person B	Shared exit gate
29-30 Oct	Import common modules and existing SPI. Freeze wiring. Run loopback and extend SPI tests.	Freeze interfaces; start arbiter unit tests and I2C engine/model. Select UART branch.	Single repository, clock/reset/record contract and UART route agreed.
31 Oct-2 Nov	ACL2 identity read, config and XYZ burst; device model and interrupt capture.	HYGRO identity and conversion/read FSM; priority/aging tests; minimal PS UART print.	Both sensor paths work independently; simulated policy decisions pass.
3-5 Nov	Record fan-out, source queues and top-level shell. Static OLED text.	Dispatcher, pending logic, AXI record FIFO/IRQ and synthetic UART record output.	Synthetic record traverses both outputs; sensor transactions arbitrate correctly.
6-7 Nov	Wire ACL2/live records into hub; add ILA probes and acquisition counters.	Wire HYGRO and PS bridge; check configuration and timeout recovery.	Both real sources produce tagged UART data. No deadlock on absent-device test.
8-11 Nov	Live OLED formatting, double buffering, metrics snapshots and hardware integration.	Output UART scheduling, synthetic traffic and automated UART log collection.	Live OLED + UART; configuration demonstrably changes grant order.
12-15 Nov	Run hardware regressions, reset tests, timing closure and nominal long run.	Run stress scoreboards, aging comparison and backpressure/fault tests.	Complete working demonstration. Freeze features on 15 Nov.
16-17 Nov	Capture hardware waveforms, timing/utilization and validate final build.	Matched experiment runs, CSV processing and comparison plots.	Measurements tied to one source commit and configuration.
18-19 Nov	Architecture/wiring/report sections and demo video.	Policy/verification/results sections and slides.	Both review report; clean rebuild and full rehearsal.
20 Nov	Demonstrate and submit tested build.	Demonstrate and submit tested build.	Final submission with backups.

Rules that prevent one person blocking the other

- A provides synthetic completed records before ACL2 hardware is ready. B provides a mock dispatcher/sink before AXI and I2C are finished.
- Both write a small unit test with every module, then merge tested changes daily. The other person reviews signal timing at the boundary.
- Reserve board slots: A for SPI/OLED, B for I2C/PS, then joint integration. Only one person edits the XDC and top-level wiring at a time.
- At the daily 15-minute check-in, show one result, identify one blocker and agree the next handoff. Use issues tagged A, B or joint.

IMPLEMENTATION PLAYBOOK / 08

Bring-up checklist and test files

Hardware sequence: follow it in this order

1. Board: clock/reset/LED and selected UART route.
2. ACL2: CS held across 0x0B + 0x00 + dummy byte; expected device ID 0xAD, then configuration and six-byte XYZ burst from 0x0E. Only then enable data-ready interrupt. [1,4]
3. HYGRO: verify IDs, set combined conversion, trigger, wait, then read four bytes. Use timer events, never a fictional DRDYn input. [2,5]
4. OLED: correct power/reset sequence, static text, then live values. [3]
5. Full system: UART acquisition first, then add display, configuration and measurements.

Proposed real-device settings: 100 MHz PL, 1 MHz ACL2/OLED SPI, 100 kHz I2C, ACL2 100 Hz, conservative 20 ms HYGRO conversion wait with bounded timed retry, readings spaced by at least one second, OLED 5 Hz. Treat these as starting settings, not measured limits. ACL2 IRQ clear may be delayed after data read; rearm without creating duplicate services. [2,4]

Required file under sim/	Purpose	By
tb_common.sv	Reuse supplied common tests; rerun in your Vivado release.	A
tb_spi_master.sv	Extend existing loopback: independent MISO data, byte boundaries, reset, timeout.	A
models/adxl362_model.sv; tb_acl2_service.sv	Register transactions, continuous CS, XYZ order, interrupt set/clear and delayed deassertion.	A
tb_i2c_master.sv	START/STOP, address bits, ACK/NACK, last-byte NACK and error termination.	B
models/hdc1080_model.sv; tb_hygro_service.sv	Conversion wait, no interrupt pin, not-ready NACK, no accidental conversion restart.	B
tb_arbiter.sv	Priority, ties, overdue selection, saturation, arrivals during service and bounded ownership.	B
tb_outputs.sv; models/oled_monitor.sv	Whole UART records, FIFO overflow behavior, OLED initialization and immutable frames.	A
tb_axi_regs.sv	AXI channel stalls, strobes, stable record peek, exactly one pop and interrupt clearing.	B
tb_system_stress.sv	Integrated hub, device models, simultaneous events, bursts, faults, backpressure and scoreboard.	A+B

Define pass/fail, not only waveforms

Each accepted transaction must produce exactly one correctly routed response. Successful records remain ordered per source; any loss must match a specified counter. No multiple grant owners, empty pops or writes beyond full. Add a watchdog to every testbench. Freeze seeds and expected results for reproducible random tests. Simulation may shorten waits, but hardware must restore real-device timing.

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Build files, measurements and completion checks

Supporting file	Why it is needed	By
README.md; .gitignore	Exact build/tool version, ownership, hardware setup; excludes caches.	A
docs/interfaces.md; register_map.md	Frozen module/record/AXI contract used by both developers.	B
docs/wiring.md; constraints/zedboard_pmods.xdc	Verified header-to-package-pin mapping, voltage standard and timing constraints. Never guess pins.	A
docs/test_plan.md; demo.md	Pass criteria, test modes and reproducible demonstration script.	A+B
scripts/create_project.tcl; create_bd.tcl; build.tcl	Recreate project, Zynq/AXI connections, IP packaging and implementation.	A+B
scripts/run_sim.tcl	Runs unit tests and system regression with consistent source order.	B
scripts/collect_uart.py; analyze_results.py	Captures logs, converts to CSV and plots policy comparisons.	B
results/run_manifest.csv; raw logs and plots	Commit, build/configuration, seed, clock, policy and run duration for each result.	B

Measure the scheduler separately from the sensor

Transaction wait: eligible-pending to accepted grant. **Acquisition response:** initial event to completed sensor record, including HYGRO conversion. **Output delay:** completed record to output acceptance/completion. Also log per-source completed count, error count, maximum wait, overflow and coalescing. Do not call conversion time arbitration latency.

Compare aging OFF and ON with identical offered traffic, seeds, service times, buffer sizes and run duration. Real-sensor trials validate hardware. Synthetic bounded-service trials demonstrate starvation/aging reproducibly. Label them separately; do not force HYGRO into unrealistically rapid reads. Equal grant counts are only a fairness target for comparable demands. Finite tests demonstrate observed waiting; they do not prove an unconditional bound under unlimited sink stalls or overload.

Use 64-bit sums for aggregate waits and coherent snapshots. At 100 MHz a 32-bit timestamp wraps in about 42.95 s: use modulo subtraction only for shorter intervals, and a wider timer/epoch for long runs. Retain raw logs, not just screenshots.

Submission gate

- Both live sensors function, OLED is stable, UART lines are intact, and reset reliably recovers the design.
- Fixed priority and aging are configurable; the same stress trace shows lower-priority progress with aging enabled.
- All unit/system tests pass; final implemented design meets its timing constraints and has no unexplained counter losses.
- Save matching source commit, XDC, scripts, hardware handoff, bitstream, ELF, reports, logs, plots, report, slides and demonstration video.

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Start now: first actions, fallback and references

Person A: next concrete actions

1. Create the shared repository and add rtl/common plus the existing spi_master.
2. Run tb_common and the supplied SPI loopback locally; add a timeout and independent slave response test.
3. Implement acl2_service with CS held low over complete transactions.
4. Verify a device-ID read in the model, then on hardware with an id_ok indicator.
5. Add XYZ acquisition, then synchronize and service the real ACL2 interrupt.

Person B: next concrete actions

1. Write interfaces.md together with A; freeze the existing record format and command handshakes.
2. Implement and test the reusable arbiter using synthetic requesters; aging may be tested before any sensor works.
3. Start I2C byte/transaction tests and HDC1080 model, then the timed HYGRO service FSM.
4. Prove the selected UART route early: PS UART print, AXI scratch register, synthetic record read/pop and IRQ.
5. Connect the dispatcher only after each endpoint obeys the agreed handshake.

Reduce scope intelligently if behind

Keep real sensor interfaces, configurable arbitration/aging, tests, UART results and a simple OLED display. Use four plain text rows; reuse validated IP/templates for AXI and standard infrastructure. Defer Linux, DMA, fancy graphics, runtime command parsing, a third requester and sharing physical SPI wires. Do not remove the starvation test or replace missing sensor interrupts with fictional pins.

Wiring and ownership reminders

A owns the board XDC; B reviews it against the schematic. Proposed connector allocation is ACL2 on JA, HYGRO on JB and OLED on JC, subject to actual board revision and physical fit. Confirm 3.3 V, ground, orientation and PL connectivity. ACL2 requires INT1 in addition to the SPI signals; OLED requires its control/power pins. Check all package-pin mappings against official documents.

Manufacturer references

[1] Digilent PMOD ACL2 reference manual

[2] Digilent PMOD HYGRO reference manual

[3] Digilent PMOD OLED reference manual

[4] Analog Devices ADXL362 datasheet

[5] Texas Instruments HDC1080 datasheet

[6] ZedBoard Hardware User Guide v2.2

Manufacturer references govern device behavior. Module boundaries, ownership, timing defaults, project register offsets and schedule are proposed choices. The primary baseline is PL sensor/arbitration logic with a minimal PS UART bridge; the strict PL-only UART branch is an alternative, not a second task to implement.